

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	14 July 2026
Project Name	Food Ordering Behaviour and Customer Trends
Team Member	Bhoomi Jain
Maximum Marks	4 Marks

Technical Architecture

The diagrammatic data ingestion pipeline runs sequentially across local processing nodes and third-party hosting architectures:

Architecture Flow Path:
Food Ordering Dataset (CSV, ~50,000 records) → Data Cleaning & Preparation (Python) → Tableau Desktop Public (Visualization Build) → Interactive Dashboard + Interactive Story (Tableau Public) → Flask Web Application (Python) → Web Browser (Embedded Dashboard View) → Target End Users (Restaurant Manager | Business Analyst | Decision Maker).

TABLE-1: COMPONENTS & TECHNOLOGIES

S.No	Component	Description	Technology
1	User Interface	How users interact with the analytics application, viewing dashboards through a web browser. Web-based Dashboard embedded via the Flask application.	HTML5, CSS3, JavaScript
2	Application Logic-1	Handles HTTP routing, renders web pages, and embeds the Tableau dashboard within the Flask application.	Python, Flask
3	Application Logic-2	Cleans and prepares the raw CSV dataset prior to visualization (handling missing values, formatting fields, etc.)	Python
4	Application Logic-3	Integrates the Tableau dashboard and story into the web application using an embedded view.	Tableau Public (Embedded View), HTML5
5	Database	This project does not use a traditional database; data is sourced directly from a CSV file.	Not Applicable for this project
6	Cloud Database	Not Applicable for this project.	Not Applicable for this project
7	File Storage	Storage of the raw and cleaned food ordering dataset used to build the dashboard.	Local File System (CSV)
8	External API-1	Not Applicable for this project.	Not Applicable for this project

S.No	Component	Description	Technology
9	External API-2	Not Applicable for this project.	Not Applicable for this project
10	Machine Learning Model	Not Applicable for this project.	Not Applicable for this project
11	Infrastructure (Server / Cloud)	Application Deployment on Local System. Local Server Configuration: Flask development server run on Windows/macOS with Python 3. Version Control: Source code maintained using Git and hosted on GitHub (no cloud server used).	Local Machine, Tableau Public (Dashboard Hosting)

TABLE-2: APPLICATION CHARACTERISTICS

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Open-source frameworks and libraries used to build the application.	Flask (Python), HTML5, CSS3, JavaScript
2	Security Implementations	Public Tableau dashboard with controlled repository access through GitHub; no sensitive or personal data is exposed in the published dashboard.	GitHub repository access control
3	Scalable Architecture	New dashboards, datasets, and visualizations can be added without changing the overall application architecture.	Modular Flask application structure
4	Availability	Dashboard is accessible through Tableau Public and the Flask web application without dependency on a dedicated hosting server.	Tableau Public hosting
5	Performance	Optimized Tableau visualizations and efficient dashboard loading through pre-aggregated views and filtered data extracts.	Tableau Public rendering optimization

References

- <https://c4model.com/>
- <https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>
- <https://www.ibm.com/cloud/architecture>
- <https://aws.amazon.com/architecture>
- <https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>